

## Chapter 5 - Exercises Q2 and Q7

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Group 2

### Question 2

A bootstrap sample of size  $n$  is formed by drawing  $n$  observations with replacement from the original sample. We consider one particular observation,  $j$ .

(a) On one draw, the probability of selecting observation  $j$  is  $1/n$ . Therefore, the probability that the first bootstrap observation is not  $j$  is

$$1 - \frac{1}{n} = \frac{n-1}{n}.$$

(b) Since sampling is with replacement, the second draw has the same probability. Thus,

$$P(\text{second draw is not } j) = 1 - \frac{1}{n}.$$

(c) For observation  $j$  to be absent from the whole bootstrap sample, it must not be selected in any of the  $n$  draws. Therefore,

$$P(j \text{ is not selected}) = \left(1 - \frac{1}{n}\right)^n.$$

So the probability that  $j$  is included at least once is

$$P(j \text{ is included}) = 1 - \left(1 - \frac{1}{n}\right)^n.$$

(d) For  $n = 5$ ,

$$1 - \left(\frac{4}{5}\right)^5 = 0.67232.$$

So the probability is about **0.6723**, or **67.23%**.

(e) For  $n = 100$ ,

$$1 - \left(\frac{99}{100}\right)^{100} \approx 0.63397.$$

So the probability is about **63.40%**.

(f) For  $n = 10,000$ ,

$$1 - \left(\frac{9999}{10000}\right)^{10000} \approx 0.63214.$$

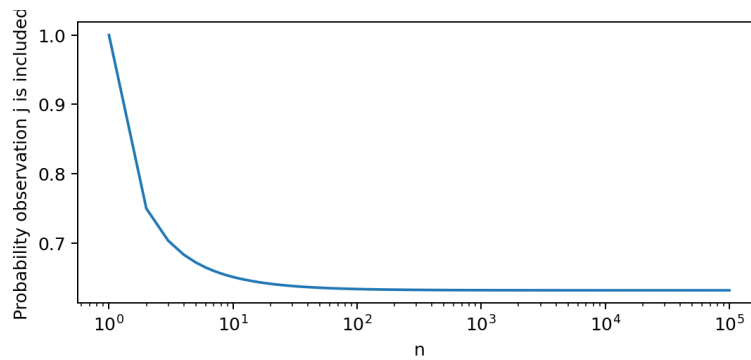
So the probability is about **63.21%**.

(g) The probability can be plotted for values of  $n$  from 1 to 100,000 using the following code.

```
import numpy as np
import matplotlib.pyplot as plt
```

```
n = np.arange(1, 100001)
p = 1 - (1 - 1/n)**n
```

```
plt.plot(n, p)
plt.xscale('log')
plt.xlabel('n')
plt.ylabel('Probability')
plt.show()
```



As  $n$  increases, the probability approaches

$$1 - e^{-1} \approx 0.6321.$$

This shows that a particular observation has about a 63.2% chance of appearing at least once in a large bootstrap sample.

**(h)** For  $n = 100$ , we can estimate the same probability by simulation.

```
import numpy as np

rng = np.random.default_rng(10)
store = np.empty(10000)

for i in range(10000):
    sample = rng.choice(100, size=100, replace=True)
    store[i] = np.sum(sample == 4) > 0

np.mean(store)
```

Using the seed above, the result is about **0.6362**. This is close to the theoretical value of 0.6340. The small difference is expected because the simulation is based on a finite number of repetitions.

### Question 7

The Weekly data set is used to predict Direction using Lag1 and Lag2. Leave-one-out cross-validation (LOOCV) is then used to estimate the test error.

(a) Fit a logistic regression model using Lag1 and Lag2.

```
import statsmodels.api as sm
from ISLP import load_data

Weekly = load_data('Weekly')
y = (Weekly['Direction'] == 'Up').astype(int)
X = sm.add_constant(Weekly[['Lag1', 'Lag2']])

model = sm.GLM(y, X, family=sm.families.Binomial()).fit()
print(model.summary())
```

The fitted model is approximately

$$\log \left( \frac{P(Up)}{1 - P(Up)} \right) = 0.2212 - 0.0387 \text{Lag1} + 0.0603 \text{Lag2}.$$

The  $p$ -value for Lag1 is about 0.140, while the  $p$ -value for Lag2 is about 0.023. Therefore, Lag2 is statistically significant at the 5% level, while Lag1 is not.

(b) Fit the model again using all observations except the first one.

```
model2 = sm.GLM(
    y.iloc[1:],
    X.iloc[1:],
    family=sm.families.Binomial()
).fit()
```

The estimated coefficients are approximately 0.2232 for the intercept, -0.0384 for Lag1, and 0.0608 for Lag2. They are very close to the results in part (a), which is reasonable because only one observation was removed.

(c) For the first observation, the predicted probability of an upward market movement is about

$$\hat{P}(Up) = 0.5714.$$

Since this is greater than 0.5, the model predicts **Up**. The actual direction for the first observation is **Down**, so the first observation is misclassified.

(d) The following loop performs LOOCV. Each observation is left out once, the model is fitted to the remaining observations, and the left-out observation is predicted.

```
import numpy as np

n = len(Weekly)
errors = np.zeros(n)

for i in range(n):
    train = np.ones(n, dtype=bool)
    train[i] = False

    fit = sm.GLM(
        y.iloc[train],
        X.iloc[train],
        family=sm.families.Binomial()
    ).fit()

    p = fit.predict(X.iloc[[i]]).iloc[0]
    pred = 'Up' if p > 0.5 else 'Down'
    errors[i] = int(pred != Weekly['Direction'].iloc[i])
```

There are **490 misclassified observations** out of 1,089.

(e) The LOOCV test error is the average of the error values:

$$\frac{490}{1089} \approx 0.450.$$

Therefore, the estimated test error is about **45.0%**, which means the model correctly classifies about **55.0%** of the observations. This suggests that Lag1 and Lag2 alone do not give very strong predictions of weekly market direction.